

Bioinformatics R Project Report

Introduction

This project demonstrates the application of R programming in solving practical bioinformatics problems. The project is divided into three main sections:

1. Transcriptomic Pipeline & Quality Control
2. Viral Variant & Sequence Integrity Analysis
3. Proteomics Sparsity & Fold-Change Analysis

The project applies several important bioinformatics concepts including data preprocessing, normalization, missing value recovery, sequence alignment, transcription, translation, GC content analysis, protein scaling, and fold-change visualization.

The following libraries were used in the project:

- Biostrings → biological sequence analysis
- tidyverse → data manipulation and visualization
- pwalgn → sequence alignment
- ggplot2 → graphical visualization

Problem 1: Transcriptomic Pipeline & Quality Control

Objective

The objective of this section is to simulate RNA-Seq count data and perform preprocessing and normalization steps commonly used in transcriptomic analysis.

Step 1: Count Matrix Generation

A count matrix containing expression values for 5,000 genes across 6 samples was generated using the `rpois()` function.

The samples were divided into:

- 3 Control samples
- 3 Treated samples

Important Functions

- `set.seed(123)` ensures reproducibility.

- `matrix()` creates the count matrix.
- `rownames()` assigns gene names.
- `colnames()` assigns sample names.

Biological Meaning

In transcriptomics, count matrices represent the number of sequencing reads mapped to each gene. Genes with higher counts are considered more highly expressed.

Step 2: Noise Filtration

Genes with total counts less than 15 across all samples were removed.

Formula

CPM filtering condition:

Total Gene Count > 15

Biological Importance

Genes with extremely low counts are usually considered noise and may negatively affect downstream analysis.

Step 3: Missing Value Recovery

Step A: Adding Missing Values

Five random cells were converted into NA values to simulate missing experimental data.

Step B: Imputation

Missing values were replaced using the mean value of other replicates within the same experimental group.

Mean Formula

Mean = Sum of values / Number of values

Biological Importance

Using replicate averages preserves biological consistency while minimizing the impact of missing values.

Step 4: CPM Normalization

Counts Per Million (CPM) normalization was manually calculated using the following formula:

$$\text{CPM} = (\text{Gene Count} / \text{Total Sample Counts}) \times 1,000,000$$

Purpose

CPM normalization corrects for differences in sequencing depth between samples, allowing fair comparison of gene expression values.

Step 5: Data Visualization

Two boxplots were generated:

1. Log2-transformed raw counts
2. CPM-normalized counts

Purpose of Visualization

The boxplots help evaluate:

- Distribution consistency
- Presence of outliers
- Effectiveness of normalization

Important Parameter

las = 2 rotates axis labels vertically for better readability.

Problem 2: Viral Variant & Sequence Integrity

Objective

This section analyzes viral DNA sequences to identify mutations and evaluate their biological effects.

Step 1: Global Sequence Alignment

Two DNA sequences were analyzed:

- Wild Type sequence
- Variant sequence

Global alignment was performed using `pairwiseAlignment()`.

Biological Importance

Sequence alignment identifies mutations, insertions, deletions, and conserved regions between viral strains.

Step 2: Transcription

The DNA sequence was converted into RNA by replacing thymine (T) with uracil (U).

Central Dogma

DNA → RNA → Protein

Biological Importance

Transcription is the process of converting DNA information into RNA molecules.

Step 3: Translation

The variant DNA sequence was translated into a protein sequence using `translate()`.

Biological Importance

Translation converts nucleotide codons into amino acids, producing proteins required for cellular functions.

Step 4: GC Content Analysis

GC content percentage was calculated using the following formula:

$$\text{GC\%} = ((G + C) / \text{Total Bases}) \times 100$$

Biological Importance

GC content affects:

- DNA stability
- Melting temperature
- Gene regulation

Sequences with higher GC content are generally more stable.

Step 5: Mutation Report

The translated protein sequence was checked for premature stop codons (*).

Interpretation

- Presence of internal stop codons → truncated protein
- Absence of stop codons → full-length protein

Biological Importance

Premature stop codons may produce nonfunctional proteins and are associated with genetic defects.

Problem 3: Proteomics Sparsity & Fold-Change Analysis

Objective

The purpose of this section is to analyze proteomics intensity data while handling missing values and comparing protein abundance between two groups.

Step 1: Proteomics Intensity Matrix

A random protein intensity matrix was generated containing:

- 800 proteins
- 6 samples
- Missing values (NA)

Biological Meaning

Proteomics experiments often contain missing values due to low protein abundance or technical limitations.

Step 2: Valid Value Filtering

Proteins were retained only if detected in at least:

- 2 out of 3 replicates in Group A
OR
- 2 out of 3 replicates in Group B

Biological Importance

This filtering removes unreliable proteins with insufficient experimental evidence.

Step 3: Low-Abundance Imputation

Missing values were replaced using:

Replacement Value = Minimum Detected Value / 5

Biological Importance

This approach assumes missing proteins are present at very low abundance rather than completely absent.

Step 4: Z-score Scaling

Protein intensities were standardized using the Z-score formula:

$$Z = (x - \mu) / \sigma$$

Where:

- x = observed value
- μ = mean
- σ = standard deviation

Purpose

Z-score normalization:

- Centers data around zero
 - Standardizes variance
 - Improves comparison between proteins
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Step 5: Fold Change Calculation

Fold change was calculated using:

Fold Change = Mean(Group B) / Mean(Group A)

Interpretation

- Fold Change > 1 → protein upregulated in Group B
 - Fold Change < 1 → protein downregulated in Group B
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Step 6: Visualization

A scatter plot was generated using ggplot2.

Plot Features

- X-axis → Group A mean intensity
- Y-axis → Group B mean intensity
- Point color → Fold change

Biological Importance

The scatter plot visually highlights proteins with strong expression differences between the two groups.

Conclusion

This project demonstrates several important bioinformatics workflows using R programming.

The project successfully implemented:

- Transcriptomic preprocessing
- Missing value handling
- CPM normalization
- Sequence alignment
- Transcription and translation
- GC content analysis
- Proteomics filtering
- Z-score normalization
- Fold-change analysis
- Data visualization

Overall, the project provides practical experience in computational biology and demonstrates how R can be used for large-scale biological data analysis.

References

1. Bioconductor Documentation
2. Biostrings Package Manual
3. ggplot2 Documentation

4. RNA-Seq Normalization Concepts
5. Molecular Biology Principles (Transcription & Translation)